CHAPTER 7

Electrochemical signaling mechanism in cardiac muscle

7.1 Introduction

Cardiac muscle fibers have a striated appearance. Muscle fibers contain many myofibrils and contract by means of the sliding filament mechanism like skeletal muscle. In contrast to skeletal muscle, cardiac muscle fibers are shorter and interconnected which makes coordinate contractions possible without having individual innervation. The cyclic release of stored Ca^{2+} from the SR (sarcoplasmic reticulum) produces the tone of the muscle and was thought to be a basic source of rhythmic cardiac automaticity. Cardiac muscle has innervation of postganglionic parasympathetic and sympathetic nerves which plays an important role in the modulation of the automaticity.

If electrical stimulations are added to the vagus nerve which is the parasympathetic nerve, the heart slows down beating. The experiment that Dr. Loewe did in 1921 was to connect two frog hearts via the small glass container full of oxygenated Ringer's solution and stimulate the nerve from the head to one heart. The solution perfusing the first heart A was applied to the second heart B via the glass container. When he stimulated the nerve to frog heart A, it slowed the beating, but a little while later heart B also showed the reduced beating. The experiment suggested that there was a substance diffusing from heart A to heart B. The diffusible substance that slowed heart A and heart B beating was called the vagus stuff. We now know that the vagus stuff is ACh. Adding ACh in the same solution also reduces heart beating. It acts on acetylcholine receptors called the muscarinic acetylcholine receptor which is entirely different from the nicotinic acetylcholine receptors on the membrane of skeletal muscle fiber at the neuromuscular junction. These muscarinic acetylcholine receptors are activated by the mushroom toxin, muscarine.

If the released ACh from the stimulated vagus nerve acted on the ACh-gated Na^{+} channel in the sarcolemma of the muscle fiber, the postsynaptic depolarization would generate the action potential, causing Ca^{2+} discharge

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from the sarcoplasmic reticulum with a resultant contraction of the cardiac muscle. However, ACh induces slowing of the heart rate and a negative inotropic effect on hearts, i.e., a decrease in the force of muscular contractions. Today, we know muscarinic acetylcholine receptors in the cardiac muscle are G_i protein-coupled receptors which are called muscarinic M_2 cholinergic receptors in pacemaker cells. The receptors are on the polar region of columnar proteins embedded in the cell membrane to detect ACh outside the cell. When ACh from the vagus nerve binds selectively to the receptor in the cardiac muscle, the conformational change activates dissociation of α subunit from the G_i protein that can inhibit adenylyl cyclase activity and operate negative action by means of a decrease in intracellular second messenger, cyclic AMP [1]. A decrease in cyclic AMP level causes the inhibition of hyperpolarization-activated cyclic nucleotide-gated (HCN) cation channels (see Section 7.6) and a decrease in cytosolic Ca^{2+}, which make the heart less excitable. In addition, muscarinic M_2 cholinergic receptors (G_q protein-coupled receptors) dissociate $\beta\gamma$ subunit to couple to muscarinic potassium channels to increase in outward currents of potassium [2], which cause hyperpolarization in a resting state called maximum diastolic potential and the resultant slowing of the heart rate. The decrease in cytosolic Ca^{2+} and slowing of the heart rate may account for the negative effects of ACh. We will discuss it in detail in this chapter.

On the other hand, stimulation of β_1 adrenergic receptors (Gs protein-coupled) mediates an activation of adenylyl cyclase and a resultant increase in cyclic AMP in the cardiac muscle. Stimulation of sympathetic nerve stimulates β_1 adrenergic receptors, which elicits an increase in cyclic AMP. The cAMP increases cytosolic Ca^{2+} level [3], which is followed by the activation of the Na^+-Ca^{2+} exchanger, resulting in an influx of net cations (3 Na^+ in, 1 Ca^{2+} out) thereby causing cellular depolarization. And together with an inhibition of inwardly rectifying K^+ currents [4] and activation of HCN channels, an increase in Ca^{2+} is implicated in making cardiac muscles to be more excitable. Thus, the sympathetic nervous system can modulate the excitability of the pacemaker cells through the second messenger cAMP. Hence, we call the cAMP the modulator of cardiac automaticity which may increase the frequency of action potential in the sinoatrial (SA) node pacemaker cells.

An action potential initiated at the SA node pacemaker cells propagates to the atrioventricular (AV) node via the internodal pathways, and to ventricular contractile cells via the His-Purkinje cells and causes rhythmic coordinate contraction of the heart muscle. Nerve impulses cannot initiate the rhythmic contraction of cardiac muscle, but they can modulate the contraction rate and its intensity. The membranes of adjacent cardiac muscle fibers interlock each other. The impulse can pass directly from cell to cell through the gap junctions. Thus, heart muscle fibers do rhythmic coordinate contractions without having individual innervation.

7.2 Function of inwardly rectifying K^+ channels in the cardiac myocytes

Inwardly rectifying potassium (K_{ir}, IRK) channels are activated by the membrane phopholipid, i.e., phosphatidylinositol 4,5-bisphosphate (PIP_2) and they pass positive potassium ions more easily in the inward direction into the cell. The inward rectification was discovered in cardiac muscle cells in the 1960s [5] and in 1970 in skeletal muscle cells [6]. In cardiac myocytes, the conductance of IRK channels dominates the resting conductance of leak K^+ channels. IRK channel attenuates outward current below −70 mV essential for rapid restoration of the resting potential from hyperpolarization probably by means of plugging the channel pore. And it restores an outward current between −70 and −55 mV essential for stabilization of the resting membrane potential probably by means of removal of the plug from the channel pore. However, in the operating potential range from −55 to +30 mV, IRK current in the outward direction decreases probably through closing the channel pore again, allowing more potassium ions to stay in the cell and delaying the repolarization (Fig. 7.1). IRK current is thought to be involved in determining the shape of the cardiac action potential through inducing rapid final finish of hyperpolarization, setting the resting potential, and permitting the plateau phase of action potential [7]. In addition, IRK may play an important role in removing small irregular depolarizations during the diastole and protecting the heart from premature contractions (see Section 10.2).

The phenomenon of inward rectification of IRK channels is probably the result of high-affinity block by endogenous polyamines, namely spermine, as well as magnesium ions, that plug the channel pore at both above −55 mV [8] and below −70 mV, resulting in a decrease in outward currents. This voltage-dependent block by polyamines will result in attenuated repolarization.

IRK channels differ from voltage-gated potassium channels since the voltage-gated potassium channels carry outward potassium currents in their operating voltage range and are responsible for repolarizing the exciting cells

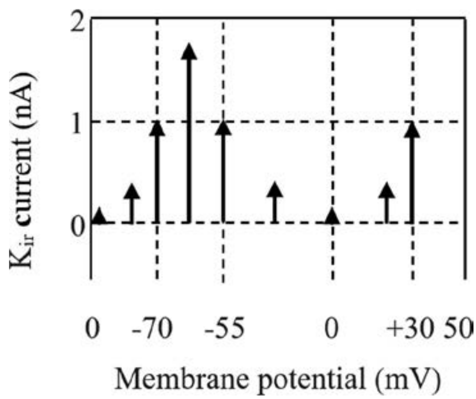

Fig. 7.1 IRK current-membrane potential relationship of cardiac muscle fiber. IRK channel attenuates outward current below −70 mV essential for rapid restoration of the resting potential from hyperpolarization probably by means of plugging the channel pore and increases an outward current between −70 and −55 mV essential for stabilization of the resting membrane potential probably by means of removal of the plug from the channel pore. However, in the operating potential range from −55 to +30 mV, IRK current in the outward direction decreases probably through closing the channel pore again, allowing more potassium ions to stay in the cell and delaying the repolarization.

whereas IRK channels play an important role in setting the resting potential, permitting the plateau phase, inducing rapid final finish of hyperpolarization, and thus preventing the heart from arrhythmias. An increase in cyclic AMP level as seen in sympathomimetic stimulation causes inhibition of inwardly rectifying K^+ currents [4] along with an increase in Ca^{2+} in the myocardium, which may increase an incident of premature contractions (see Section 10.3).

7.3 G protein-coupled receptors in the heart

G protein-coupled receptors (GPCR) are transmembrane proteins that detect ligands outside the cell and activate cellular responses by means of signal transduction pathways. When a ligand binds to the GPCR, it causes a conformational change in the GPCR, which allows it to act as a guanine nucleotide exchange factor. The GPCR activates the coupled G protein by exchanging GDP bound to the G protein for GTP. The G protein's α subunit together with the GTP, then dissociates from the βγ subunits and further affects intracellular target enzymes such as adenylate cyclase or phospholipase C directly depending on the α subunit type.

(1) Signal transduction pathways via G_α: (a) cAMP signal pathway is involved in β_1 adrenergic receptors and muscarinic M_2 receptors in cardiac muscle cells, and β_2 adrenergic receptors in the smooth muscles of coronary blood vessels. β_1 and β_2 adrenergic receptors are G_s-coupled receptors. Muscarinic M_2 receptors are G_i-coupled receptors. The effector of dissociated α subunit of G_s and α subunit of G_i is the cAMP-producing enzyme, adenylate cyclase. Stimulation of G_s-coupled receptors activates adenylate cyclase, and stimulation of G_i-coupled receptors inactivate adenylate cyclase. (b) The phosphatidylinositol signal pathway is involved in α_1 adrenergic receptors that is G_q-coupled receptors in coronary blood vessels. The effector of dissociated α subunit of G_q ptotein is the inositol triphosphate-producing enzyme, phospholipase C. The major second messenger, inositol triphosphate (IP_3) diffuses inside the cell, where it binds to its receptor, which is on the IP_3-gated Ca^{2+} channel located in the endoplasmic reticulum (ER) (see Section 8.5).

(2) Signal transduction pathways via $G_{\beta\gamma}$: G protein-gated inward-rectifying potassium (GIRK, IKACh) channel located in the sinoatrial (SA) node and atria in the heart plays a major role in the regulation of heart rate [9]. Activation of the IKACh channels begins with the release of ACh from the vagus nerve to pacemaker cells in the heart. ACh binds to the M_2 muscarinic receptors (G_i-coupled), which promotes the dissociation of $\beta\gamma$-complex from α subunit [10]. The dissociated $\beta\gamma$-complex binds directly and specifically to the IKACh channel through interactions with both the GIRK1 and GIRK4 subunits. (The IKACh channel is composed of two homologous GIRK channel subunits: GIRK1 and GIRK4.) Once the plug in IKACh channel is removed by binding to the $\beta\gamma$-complex, K^+ ions can flow out of the pacemaker cells causing hyperpolarization in a resting state during the diastole [11]. In its hyperpolarized state, action potentials cannot be generated as quickly, which slows the heart rate.

7.4 Ryanodine receptor 2 channels

Ryanodine receptor 2 (RyR2) channel is Ca^{2+}-gated Ca^{2+} channel found primarily in cardiac muscle. The RyR2 channel located in the sarcoplasmic reticulum (SR) is leaky even at rest and supplies Ca^{2+} to the cardiomyocytes to depolarize rhythmically. To enable cardiac muscle contraction, voltage-gated L-type calcium channels in the plasma membrane must be opened by the voltage above the threshold, and the calcium influx must raise an action potential which causes dihydropyridine (DHP) Ca^{2+} channels to open and allows calcium ions to bind to RyR2 located on the SR. This binding causes the massive release of calcium through RyR2 from the SR into the cytosol, where it binds to the C domain of troponin, which uncovers myosin-binding sites on actin causing the formation of the cross-linkages between actin and a myosin head with the high energy phosphate, and the successive conformational change of myosin head through the removal of the phosphate causes sliding of thin filament on the thick filament. The hydrolysis of ATP by the actin activated-myosin ATPase in the myosin head causes the myosin head to bind to the high-energy phosphate in the presence of Ca^{2+} and a recocking of the myosin head to ready for rebinding to actin, which enables the cyclic cardiac muscle contractions (see Section 8.4).

Phosphorylation of RyR2 by activated protein kinase A through sympathomimetic stimulation may make the RyR2 Ca^{2+} channel leakier [12]. In vitro, PKA can phosphorylate RyR2. However, in intact myocytes, a functional role of PKA regulation of RyR2 remains controversial [13]. β-Adrenergic receptor agonist, isoproterenol enhanced Ca^{2+} leak from the SR in intact rabbit ventricular myocytes. The Ca^{2+} leak seemed not to be related to protein kinase A activity but was thought to be mediated by CaMKII activity [14,15]. An increase in the cytosolic Ca^{2+} along with an increase in cAMP will cause cardiac muscle more excitable.

7.5 Sodium-calcium exchanger

The sodium-calcium exchanger (Na^{+}-Ca^{2+} exchanger or NCX) is a membrane protein that removes Ca^{2+} from the cells. It uses the electrochemical potential energy to allow Na^{+} to flow across the plasma membrane in exchange for the countertransport of Ca^{2+}. A single calcium ion is exported for the import of three sodium ions, which results in depolarization. The exchanger exists in many different cell types including cardiac, skeletal, smooth muscle, and neuronal cells. It is considered one of the most important cellular mechanisms for removing Ca^{2+} in a resting state [16], by which the cytoplasmic concentration of calcium ions in the cell is kept low. The exchanger does not bind very tightly to Ca^{2+}, but it can transport the ions rapidly [17]. Therefore, it is useful for ridding the cell of large amounts of Ca^{2+} in a short time, as it is needed for cardiac cells to get in a resting state quickly. Thus, the exchanger likely plays an important role in regaining the cell's normal calcium concentrations [18].

In a resting membrane potential, the Na^+-Ca^{2+} exchanger pumps Ca^{2+} out of cardiac muscle cells with the help of the large extracellular Na^+ concentration. However, during the upstroke of the action potential in a cardiac muscle fiber, there is a large influx of Na^+ ions. This depolarizes the cell and shifts the membrane potential in a positive direction. A large increase in intracellular Na^+ concentration causes the reversal action of the Na^+-Ca^{2+} exchanger, which is the pumping Na^+ ions out of the cell and Ca^{2+} ions into the cell. However, this reversal of the exchanger lasts only momentarily, and the exchanger returns to its forward direction of flow, that is, pumping the Ca^{2+} out of the cell as the cytosolic sodium concentration goes down [19].

The plasma membrane Ca^{2+} ATPase (PMCA) that exports Ca^{2+} from the cell and sarcoendoplasmic reticulum Ca^{2+} ATPase (SERCA) that imports Ca^{2+} into the SR have a much higher affinity with Ca^{2+} but their transport capacity is much lower. The Ca^{2+} ATPases are capable of effectively binding to Ca^{2+} even when their concentrations are quite low. The Na^+-Ca^{2+} exchanger complements Ca^{2+}-ATPases that have a high affinity but low capacitance, and together they are involved in muscle relaxation, maintenance of Ca^{2+} concentration in the SR in cardiac cells.

The activation of the exchanger is also implicated in the cardiac electrical conduction abnormality known as premature contraction [20]. It is thought that intracellular accumulation of Ca^{2+} causes the activation of the Na^+-Ca^{2+} exchanger as seen in sympathomimetic stimulation. The result is the influx of a net positive charge (3 Na^+ in, 1 Ca^{2+} out), thereby causing cellular depolarization. This abnormal depolarization may lead to premature contraction.

7.6 HCN cation channels and modulation of the cardiac rhythmicity

Regarding the SA node depolarization, the channel activated by hyperpolarization has been reported in the plasma membrane [21]. It is a nonselective channel for cations called hyperpolarization-activated cyclic nucleotide-gated (HCN) cation channel. They respond to bind cyclic AMP when the cell is hyperpolarized and depolarize the cell toward the threshold via cation influx called funny currents (I_f). HCN4 is the main isoform expressed in the sinoatrial node (SA node). The current through the activated HCN channels plays a key role in cardiac rhythmicity.

The HCN channels are implicated in the cardiac rhythmicity but do not likely create cyclic depolarizations in the cytosol. The abolishment of HCN channels in the mutant mice caused the failure of some impulse initiations in SA node pacemaker cells. However intermittent firing of SA node pacemaker cells is still preserved and the activation of β adrenergic receptor by isoproterenol restored the regular firing although its rate was slower than that of control mice [22]. The experimental observation suggested that depolarization caused by I_f through the HCN channels was not the basic source of the automaticity of the pacemaker cell activity and there should be a fundamental source of the automaticity elsewhere. Intracellular Ca^{2+} oscillation, i.e., the sinusoidal release of stored Ca^{2+} has been proposed as a primary source of the rhythmical SA node depolarization (see Section 5.9).

The depolarization by the HCN can be achieved in the presence of cAMP. The depolarization begins when conformational change of β_1 adrenergic receptor (G_s protein-coupled receptor) is brought by binding to norepinephrine (NE) or epinephrine (EPI), which allows it to act as a guanine nucleotide exchange factor. Stimulation of the receptor activates the G_s protein by exchanging GDP bound to the G protein for GTP. The G protein's α subunit together with the GTP then dissociates from the βγ subunits and further activates the intracellular cAMP producing enzyme, adenylate cyclase. When the HCN ion channels are under hyperpolarization, they react with the negatively charged cyclic AMP, open for nonselective positive ions, and then cause the inward positive currents (I_f). Simultaneously, a negatively charged cyclic AMP diffuses and then activates cyclic AMP-dependent protein kinase (PKA), and the liberated catalytic subunits of PKA phosphorylate the Ca^{2+} channels to open [23]. The raised depolarization by the activated HCN channels along with cAMP will induce the opening of the L-type voltage-gated Ca^{2+} channels allowing a larger Ca^{2+} influx, which increases the rate of action potentials in the node pacemaker cells.

7.7 Gap junctional conduction

Cardiac muscle fibers are short and interconnected. The membranes of adjacent cardiac muscle fibers interlock each other. The impulse can pass directly from cell to cell through the gap junctions. Bilateral movements of Ca^{2+} through gap junctional channels make cardiac muscle fibers contract coordinately. Phosphorylation of transmembrane protein connexin 43 (Cx43) is necessary to form a pore in the gap junctions for ions to pass through. The phosphorylation occurs at multiple serine residues. Cx43 is phosphorylated soon after synthesis and its phosphorylation site changes according to match the physiological event when it travels along the endoplasmic reticulum (ER) and Golgi to the plasma membrane, or when it functions as a channel in the gap junction membrane. Cx43 are classified into P_0 (nonphosphorylated isoform), P_1, and P_2 (phosphorylated isoform) according to differences in the electrophoretic mobility on sodium dodecyl sulfate-polyacrylamide gel electrophoresis. The mobility shifts probably occur through the changes in the phosphorylation site on Cx43. The phosphorylation at S365 involved in forming the P_1 isoform apparently helps its travel to or within the plasma membrane. Phosphorylation at S325, S328, and/or S330 occurs only in the gap junctional membrane and is necessary to form P_2 isoform that has a pore [24]. Under ischemic conditions, there is a loss of Cx43 at the intercalated disc as the nonphosphorylated form of Cx43 (P_0) moves to the lateral edges of the myocyte and the gap junction becomes less conductive.

The structure of gap junctional channels in the myocardium or the SA or AV nodes are not only modulated by phosphorylation of connexin 43 [25], but also glycosylation, nitrosylation, or ubiquitination [26]. Increased cytosolic Ca^{2+} concentration [27], reduced ATP concentration, and acidification [28] are known to decrease the conduction through gap junctional channels. Reduced conductivity of the gap junction with reduced ATP concentration probably occurs in the myofilament Ca^{2+} sensitization [29] and cardiac ischemia, which may decrease the ion conductivity of the tissue and lead to arrhythmias (see Section 5.2).

7.8 Action potential in the heart

Cardiac muscle has innervation of postganglionic parasympathetic and sympathetic nerves. However, nerve impulses cannot initiate the rhythmic contraction of cardiac muscle, but they can modulate the contraction rate and its intensity. Rhythmic action potentials are generated at the sinoatrial (SA) node pacemaker cells and propagate along the internodal pathways to the atrioventricular (AV) node, and farther to ventricular contractile cells via the His-Purkinje cells, and cause rhythmic coordinate contractions of the heart muscle. Although individual cardiomyocytes are separated by the plasma membrane, the impulse can propagate directly from cell to cell through the gap junctions. They permit the cells to function as if they were a syncytium. There are three types of cardiac cells specialized for impulse initiation, rapid conduction, and contraction. They are "P cells for impulse initiation in the sinoatrial (SA) and atrioventricular (AV) nodes," "the special conductive cells constituting internodal pathways from the SA node to the AV node and a pathway from the SA node to the left atria, the bundle of His, and thin filaments distributing to ventricular contractile cells called Purkinje fibers for rapid conduction," and "contractile myocardial cells for contraction."

Pale oval P cells are located centrally in the SA and AV nodes and so named since they are pale, primitive, and pacemaker cells. These cells are oval or rounded in contrast to the elongated shape of other myocardial cells. The P cells in the SA node initiate rhythmical impulses automatically at a regular speed, 60–80 beats per minute. If the SA node stopped, the P cells in the AV node would take over firing rhythmical impulses at a slightly slower pace.

The postganglionic sympathetic fibers terminating in the SA node and the AV node play an important role in accelerating the heart rate, and some fibers distributing to the ventricular muscle fiber increase the force of muscular contractions. Postganglionic parasympathetic fibers supplying the specialized cardiac muscle cells of the SA node and AV node play an important role in slowing the heart rate, and some fibers distributing to the ventricular muscle fiber decrease the force of muscular contractions [30].

To answer the question of how rhythmical impulses in pacemaker cells are initiated automatically, the first step is to examine the wave pattern of an action potential in the pacemaker cells and compare it with those of the conductive Purkinje cells and ventricular muscle fibers (Fig. 7.2). Action potentials of the nodal pacemaker cells have a blunt curve of depolarization and repolarization, and the conduction rate of impulses is slow. The pacemaker cells are called the slow-response cells. No sharp upstroke of an action potential suggests no sodium ion influx, a large blunt upstroke of the potential probably represents slow Ca^{2+} influx through the sarcolemma, and a slow rate of repolarization suggests slow K^+ efflux. The large sinusoidal wave pattern of action potentials suggests the periodic extracellular calcium influxes. The existence of cyclic depolarizations in the cytosol and its amplification is essential to induce the Ca^{2+} influxes through the voltage-gated Ca^{2+} channels.

The thin filaments called Purkinje fibers distributing to ventricular contractile cells conduct impulses most rapidly, and the bundle of His and contractile myocardial cells also conduct impulses fast. They are called the fast-response cells. The action potential of the Purkinje fibers and ventricular

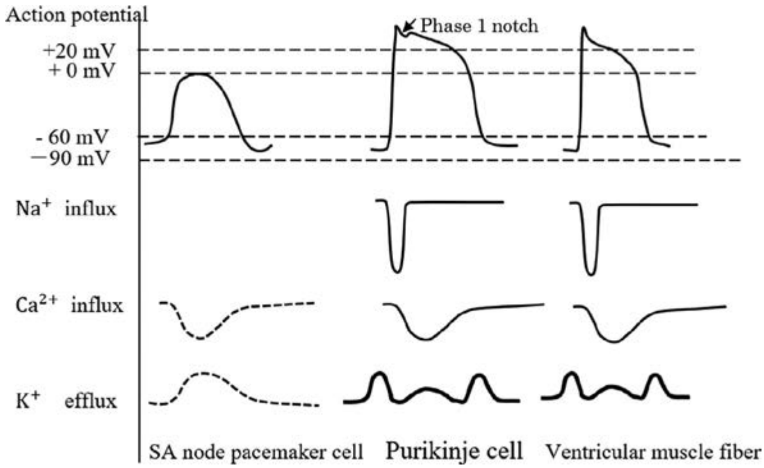

Fig. 7.2 Comparison of the wave pattern of an action potential in the pacemaker cells with those of the conductive Purkinje cells and ventricular muscle fibers. Action potentials of the nodal pacemaker cells have a blunt curve of depolarization and repolarization, and the conduction rate of impulses is slow. No sharp upstroke of an action potential suggests no sodium ion influx, the large blunt upstroke of the potential probably represents slow Ca^{2+} influx, and the slow rate of repolarization suggests slow K^{+} efflux. The large sinusoidal wave pattern of action potentials suggests the periodic extracellular calcium influxes. The sharp initial spike with the phase 1 notch from Purkinje cell may correspond to a rapid depolarization by the opening of voltage-gated Na^{+} channels followed by abrupt closure of the Na^{+} channels. The subsequent prolonged plateau is explained by a slow closure of voltage-gated Ca^{2+} channels and the attenuated K^{+} efflux. IRK channels are dominant in cardiac muscle cells and responsible for the formation of the phase 2 plateau of the action potential through the attenuated outward current of potassium ions in the operating potential range from −55 to +30 mV.

muscle fibers shows an initial sharp upstroke followed by a prolonged plateau of the potential. The sharp initial upstroke may correspond to a rapid depolarization by the opening of voltage-gated Na^{+} channels followed by abrupt closure of the Na^{+} channels. The subsequent prolonged plateau is explained by a slow closure of voltage-gated Ca^{2+} channels and attenuated K^{+} efflux. IRK channels are dominant in cardiac muscle cells and responsible for the formation of the phase 2 plateau of the action potential through the attenuated outward current of potassium ions in the operating potential range from −55 to +30 mV [7] (see Section 7.2). Voltage-gated potassium channels carry outward potassium currents and are responsible for repolarizing the exciting cells. On the other hand, IRK channels are responsible for inducing the rapid final finish of hyperpolarization (see Fig. 7.1) during the diastole.

7.9 Cyclic depolarization of the pacemaker cells and modulation by postganglionic autonomic nerves

Intracellular Ca^{2+} oscillation, i.e., the sinusoidal release of stored Ca^{2+} causes the tone of the muscle. Imaging the SA node cells showed Ca^{2+} sparks arising from the SR [31]. Instead of the HCN channels, the oscillatory release of stored Ca^{2+} from the SR was considered as the basic source of the automaticity of the pacemaker cell activity [32,33]. It was thought to be a diffusion phenomenon and justified mathematically using a model of Ca^{2+} diffusion through the leaky Ca^{2+} channel. Solving the equation of Fick's second law of diffusion, the movement of Ca^{2+} level in the channel showed damped oscillation along the straight line with a negative slope (see Section 5.9).

Using the electrochemical potential energy of Na^+ across the plasma membrane, the sodium-calcium exchanger removes Ca^{2+} from the cells during the diastole. A single calcium ion is exported for the import of three sodium ions, which likely results in amplifying depolarization caused by the oscillatory release of Ca^{2+} from the SR.

Ca^{2+}-activated K^+ channels were found to be a critical player in continuous rhythmic pacemaker activity derived from human embryonic stem cells. [34]. During the last phase of the systolic depolarization, enough cytosolic Ca^{2+} level may activate Ca^{2+}-activated outward K^+ currents which contribute to hyperpolarization. Then hyperpolarization activates HCN channels and raises I_f if there is enough cyclic AMP in the cell. IRK channels work for inducing a rapid final finish of the hyperpolarization and setting the resting potential to get ready for the next depolarization.

The rate of oscillatory depolarization in the SA node pacemaker cells is modulated by postganglionic parasympathetic and sympathetic nerves.

(1) *Slowing rhythmic impulses*: Activation of the IKACh channels begins with a release of ACh from the vagus nerve supplying the specialized myocardial tissue of the SA and AV nodes to pacemaker cells. ACh binds to the M_2 muscarinic receptors (G_i-coupled receptors), which promotes the dissociation of the βγ-complex from the α subunit [10] (see Section 7.3). Once the plug of the IKACh channel is removed by binding to the βγ-complex, more K^+ ions flow out of the pacemaker cells causing hyperpolarization in a resting state during the diastole [11]. In its hyperpolarized state, action potentials cannot be generated as quickly, which slows the heart rate. On the other hand, the dissociated α subunit can inhibit adenylate cyclase activity (Fig. 7.3) and operate negative action by means of a decrease in intracellular second messenger, cyclic AMP. A decrease in cyclic AMP level causes the inactivation of HCN channels, which may slow the heart rate and a reduction of cytosolic Ca^{2+}.

(2) *Accelerating rhythmic impulses*: Activation of the HCN channels begins with a release of NE (norepinephrine) from the postganglionic sympathetic nerve terminating in the SA and AV nodes to pacemaker cells in the heart. NE binds to the β_1 adrenergic receptors (G_s-coupled), which promotes the dissociation of the α subunit. The effector of the α subunit of G_s is the cAMP-producing enzyme, adenyl cyclase (Fig. 7.3). When HCN ion channel is activated by hyperpolarization in the presence of a negatively charged cAMP, it opens for nonselective positive ions, and then causes the inward positive currents (I_f). The depolarization evoked by the HCN channels may cause a large Ca^{2+} influx through the L-type voltage-gated Ca^{2+} channel, which may raise action potentials in the node pacemaker cells. Simultaneously, a negatively charged cyclic AMP diffuses, activates cyclic AMP-dependent protein kinase (PKA), and liberated catalytic subunits of PKA phosphorylate the Ca^{2+} channels to open [3,23]. Through inactivation of inwardly rectifying K^+ currents [4], an increase in Ca^{2+}, and activation of HCN channels, an increase in cAMP by stimulation of the sympathetic nerve is implicated in making cardiac muscles to be excitable.

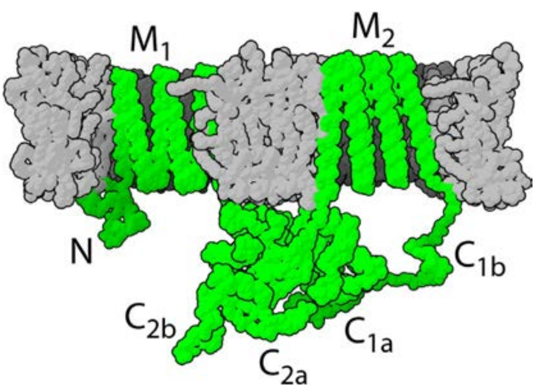

Fig. 7.3 Most adenyl cyclases are transmembrane proteins with 12 transmembrane segments and 2 cytoplasmic domains. The α subunit of G proteins associates with adenyl cyclase, which converts ATP to cAMP.

7.10 Amplification and stabilization of signals

The amplification of Ca^{2+} oscillation from the SR, rapid final finish of the hyperpolarization, setting the stable resting potential, and permitting the plateau phase of an action potential are all essential for cardiac cells to generate stable rhythmic action potentials continuously. The ectopic depolarizations probably are caused by amplified Ca^{2+} in the excitable cardiac cells with reduced conductivity (see Section 10.3).

The cardiac ryanodine (RyR2) channels are Ca^{2+}-gated Ca^{2+} channels in the SR. When leak RyR2 channels are at a resting state, the channel may allow the slow sinusoidal release of stored Ca^{2+} from the SR [35,36] to the cytosol, and then the oscillatory depolarizations may be amplified during the diastole by means of Na^{+}-Ca^{2+} exchanger (see Section 7.5). At the same time, the Ca^{2+} removal by Na^{+}-Ca^{2+} exchanger contributes to keeping continuous pacemaker activity in the SA node [37]. During the last phase of the systolic depolarization, enough cytosolic Ca^{2+} level may activate Ca^{2+}-activated outward K^{+} currents [38] which may bring hyperpolarization, and then the hyperpolarization activates HCN channels to induce the hyperpolarization-activated funny currents (I_f) [21] in the presence of enough cyclic AMP. The amplified depolarizations by Na^{+}-Ca^{2+} exchanger and I_f activates L-type Ca^{2+} channels to discharge large amounts of Ca^{2+} into the cardiac pacemaker cells.

The outward K^{+} currents through inwardly rectifying potassium (IRK) channels during the early phase of depolarization contribute to setting the stable resting potential by removing irregular small noises during the diastole. On the other hand, the attenuated outward K^{+} currents in the operating potential range stabilize the action potential by permitting the plateau phase [7] (see Section 7.2). An increase in cyclic AMP level causes increases in Ca^{2+} and I_f, and inactivation of inwardly rectifying K^{+} channels in the myocardium, which may increase an incident of arrhythmia [4,39].

7.11 Conduction between the atrium and ventricle

The thin filaments called Purkinje fibers distributing to ventricular contractile cells conduct impulses the fastest since they are thin filaments having the most voltage-gated Na^{+} channels in the sarcolemma-like neurons. The velocity of conduction is 4 m/s. The rate of conduction in the bundle of His and contractile myocardial cells is not as fast as the Purkinje fibers, and their velocities are about 1 m/s. The cells which have many voltage-gated

Na^+ channels conduct impulses fast and respond to stimuli quickly, but the refractory period is longer than slow-response cells due to the inactivation of voltage-gated sodium channels.

The conduction rate of the atrioventricular (AV) node is the slowest (0.05 m/s) since pacemaker cells in the AV node lack voltage-gated Na^+ channels where diffusion of Ca^{2+} through gap junctional channels mainly contribute to the propagation of impulses. The refractory period is shorter than fast-response cells. The slow conduction in the AV node is a prerequisite to give the atrial contractile muscle fibers a time to propel all the blood into the ventricle before the ventricles contract and protect the ventricles from supraventricular tachycardia (see Section 10.4.3). The AV node has automaticity (40–60 beats per min) with innervation of postganglionic autonomic nerve fibers. The slow conduction in the AV node can maintain the pulse rate of the ventricles in the physiological range in the case of the atrium fibrillation where the pulse in the atrium is disorganized (see Section 10.2).

7.12 Conclusions

IRK channels differ from voltage-gated potassium channels since the voltage-gated potassium channels carry outward potassium currents in their operating voltage range and are responsible for repolarizing the exciting cells whereas IRK channels play an important role in setting the resting potential, permitting the plateau phase, inducing rapid final finish of hyperpolarization, and thus preventing the heart from arrhythmias. Using the electrochemical potential energy of Na^+ across the plasma membrane, the sodium-calcium exchanger removes Ca^{2+} from the cells during the diastole. A single calcium ion is exported for the import of three sodium ions, which likely results in amplifying depolarization caused by the oscillatory release of Ca^{2+} from the SR.

The cyclic release of stored Ca^{2+} from the SR or the ER produces the tone of the muscle and was thought to be a basic source of rhythmic cardiac automaticity. Sympathomimetic stimulations activate adenylate cyclase to produce the second messenger, cAMP which activates the HCN channels to raise the pulse rate. Through an inactivation of inwardly rectifying K^+ currents [4], an increase in Ca^{2+}, and an activation of HCN channels, an increase in cAMP by stimulation of the sympathetic nerve is implicated in making cardiac muscles to be more excitable.

On the other hand, activation of the IKACh channels begins with a release of ACh from the vagus nerve supplying the SA and AV nodes.

ACh binds to the M_2 muscarinic receptors (G_i-coupled), which promotes the dissociation of the βγ-complex from the α subunit [10]. Once the plug in the IKACh channel is removed by binding to the βγ-complex, K^+ ions flow out of the pacemaker cells causing hyperpolarization in a resting state during the diastole [11]. In its hyperpolarized state, action potentials cannot be generated as quickly, which slows the heart rate. The dissociated α subunit inhibits adenylate cyclase activity and operates negative action by means of a decrease in intracellular second messenger, cyclic AMP.

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